ASTROSIGHT | PLANETARY SURFACE INTELLIGENCE

AUTONOMOUS LUNAR & MARTIAN TOPOGRAPHICAL CRATER ANALYSIS REPORT

Analyst / Researcher:	Dr. Helena Vance	**Target Body:**	**MOON**
Timestamp:	4da8fe96-27ad-4fd3-	**Resolution:**	10.0 m/px
Validation Status:	**VALID_CRATER_DETECTED**	**Craters Identified:**	10
Model Pipeline:	AstroSight-CraterNet-v2.4	**Mean Confidence:**	**93.5%**

1. TOPOGRAPHICAL CRATER DETECTIONS & SPATIAL VECTORS

ID	Feature Name	X (px)	Y (px)	Radius (px)	Diameter (km)	Confidence
crater_01	CRATER 01	400.0	224.0	176.0	3.52 km	91.2%
crater_02	CRATER 02	13.0	175.0	168.0	3.36 km	91.7%
crater_03	CRATER 03	188.0	268.0	149.0	2.98 km	92.2%
crater_04	CRATER 04	413.0	184.0	139.0	2.78 km	92.7%
crater_05	CRATER 05	389.0	279.0	135.0	2.7 km	93.2%
crater_06	CRATER 06	215.0	160.0	133.0	2.66 km	93.7%
crater_07	CRATER 07	187.0	407.0	128.0	2.56 km	94.2%
crater_08	CRATER 08	249.0	286.0	127.0	2.54 km	94.7%
crater_09	CRATER 09	177.0	151.0	126.0	2.52 km	95.2%
crater_10	CRATER 10	254.0	124.0	123.0	2.46 km	95.7%

2. EUCLIDEAN & REAL-WORLD SPATIAL DISTANCE MATRIX

Vector	Origin Crater	Target Crater	Pixel Dist (px)	Real Distance (m)	Real Distance (km)
VEC-01	CRATER 01	CRATER 02	390.09 px	3900.9 m	3.90 km
VEC-02	CRATER 01	CRATER 03	216.52 px	2165.2 m	2.17 km
VEC-03	CRATER 01	CRATER 04	42.06 px	420.6 m	0.42 km
VEC-04	CRATER 02	CRATER 03	198.18 px	1981.8 m	1.98 km
VEC-05	CRATER 02	CRATER 04	400.10 px	4001.0 m	4.00 km
VEC-06	CRATER 02	CRATER 05	390.12 px	3901.2 m	3.90 km
VEC-07	CRATER 03	CRATER 04	240.17 px	2401.7 m	2.40 km
VEC-08	CRATER 03	CRATER 05	201.30 px	2013.0 m	2.01 km
VEC-09	CRATER 03	CRATER 06	111.32 px	1113.2 m	1.11 km

VEC-10	CRATER 04	CRATER 05	97.98 px	979.9 m	0.98 km

3. SCIENTIFIC & TECHNICAL SPECIFICATIONS

Mathematical Formulation: Planar Euclidean metric d = √((x■ - x■)² + (y■ - y■)²) scaled via orthorectified ground sampling distance 10.0 m/px. Real-world distance D = d × resolution.
Sustainable Development Goal Alignment: AstroSight aligns with United Nations SDG 9 (Industry, Innovation, and Infrastructure) by advancing automated planetary remote sensing, AI spatial analytics, and space exploration research infrastructure.

Scientific Disclaimer: *AstroSight automated crater boundary coordinates and distance vectors are generated for scientific research and mission planning. Topographical scale assumes standard planar projection for small-area orbital imagery. Always cross-correlate with high-resolution LROC NAC / HiRISE elevation models.*